

Industrial IoT Platform – Node Architecture Breakdown (Tabular)

Layer	Component	Recommended Nodes	Notes
Gateway	Protocol Ingestion	3	Handles Modbus, OPC-UA, MQTT
Streaming	NATS JetStream	3	Cluster with replication
Processing	Processing Engine	3	Mapping, filtering, events
Cache	Redis Cluster	6	3 master + 3 replica
WebSocket	Real-time Fan-out	2	Client connections
Storage	ClickHouse	3	Distributed analytics DB
Control Plane	Keycloak + PostgreSQL	3	2 Keycloak + 1 DB
Observability	Prometheus + Loki + Grafana	2	Monitoring and logs

Total Logical Nodes: ~25

Optimized Deployment: ~20–21 nodes

Physical Machines Required: ~8–12 (using Kubernetes/K3s)